



Working with Labeled Volumes

A labeling session generates a groundTruthMedical Object and labeled volume data. These can be viewed and analyzed outside the app.

Navigate to the Module 3 folder in the course files and work through the **AnalyzingLabeledVolumes.mlx** script. You'll work with a completed labeled volume of the MRI data you saw in the previous video to compute properties like volume for each label.

We encourage you to work through the code in MATLAB. A static file is provided below. The code may be different than in MATLAB because interactive tools are not present in static files.

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Labels often correspond to specific structures in your volume. Your analysis may require reporting various statistics for each label.

Use `regionprops3` to measure properties for each labeled region.

The following code computes the volume of each label in the segmented volume.

```
segProps = regionprops3(segVol.Voxels, "Volume")
```

Because the input is `segVol.Voxels`, the output is in voxels, meaning the number of voxels with the corresponding ID.

Multiply the result by the real-world voxel-spacing to convert the value to mm^3 .

```
segProps.VolumeMM3 = segProps.Volume*prod(segVol.VoxelSpacing)
```

Measure Labeled Voxel Values

With the volume segmented into various regions of interest, it is also possible to make measurements of voxel intensity values by region.

Use `regionprops3` again, with the first input being the labeled voxel values, the second being the corresponding intensity voxel values, and the third specifying the property to measure.

Note: If this code returns an error, [add the course Data folder to your MATLAB Path](#).

```
% Select the T1_3T_1018 folder in the course data folder
mrifolder = "";
medvol = medicalvolume(mrifolder);
volProps = regionprops3(segVol.Voxels, medvol.Voxels, "MeanIntensity")
```

The output is in the same units as voxel intensity.

Hounsfield Units

The intensity values of CT images may be calibrated to correspond to different tissue density levels. This calibrated intensity is reported in Hounsfield Units.

The intensity values of medicalimage and medicalVolume objects are automatically rescaled to Hounsfield Units if the associated metadata properties are present.

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